Causal-Inspired Event-Token Spatio-Temporal Graph Learning for Alarm-Driven KPI Degradation Prediction in Cellular Networks

Dun Li, Hongzhi Li, Noel Crespi, Roberto Minerva, Ming Li, Wei Liang, Kuan-Ching Li

Abstract—**Cellular network operations are monitored through heterogeneous time-evolving signals, including KPI measurements, business counters, sparse alarm/fault events, cell identifiers and temporal context. Predicting future KPI degradation is operationally important but difficult because alarms are sparse, KPI responses may be delayed, physical topology is often unavailable and expert-confirmed root-cause labels are not always recorded. We study a real private cellular operations panel extracted from `510957.csv`, containing 103,983 rows, 33 cells, 3,151 hourly timestamps, 48 KPI/business columns and 137 alarm/fault columns, of which 11 are active. We propose CET-STGL, a causal-inspired event-token spatio-temporal learning pipeline that organizes KPI histories into patch tokens, sparse alarms into event tokens, cell/time context into positional features and train-only alarm-degradation associations into an event influence prior with inferred cross-cell context. The current runnable implementation evaluates an sklearn proxy rather than a neural graph backend because PyTorch is unavailable. Experiments cover 1/3/6/12-hour KPI forecasting, weak KPI degradation classification and weak root-cause candidate ranking. Results show strong KPI-history baselines, weak standalone event-only predictors and competitive CET-STGL proxy forecasting on several primary-KPI metrics, while also showing that alarm-token gains are not uniform and weak ranking is not expert root-cause validation.**

Index Terms—**Cellular networks, KPI forecasting, alarm correlation, weak root-cause ranking, spatio-temporal graph learning, causal-inspired event influence.**

I. Introduction

Cellular networks are operated through structured telemetry rather than a single homogeneous signal. A cell-level operations panel may contain continuous KPI and business counters, sparse alarm/fault events, cell identifiers, and time context. KPI histories describe load, accessibility, retainability, throughput and radio-quality dynamics; alarms mark discrete operational events; cell and time features encode persistent local behavior and periodic demand. Anticipatory mobile networking and traffic-prediction studies have shown that prediction can support proactive resource management and operational optimization [1], [2]. For cellular operations, anticipating KPI degradation is useful because it can surface risky cell states before manual inspection or customer impact.

The modeling difficulty is that these signals are heterogeneous. Pure KPI-history models can exploit strong temporal regularities, but they may ignore sparse event context that operators naturally inspect during fault handling. Pure alarm/event models, on the other hand, are often insufficient for continuous KPI prediction because alarm occurrences are rare and many KPI changes follow routine traffic or radio patterns rather than explicit faults. Prior cellular traffic work has emphasized spatial and temporal dependencies among base stations [3], [4], while alarm-correlation work has long noted that a single fault may generate many related alarms and that fault isolation is difficult in large communication networks [5], [6]. In enterprise operations data, the problem is harder still: physical topology may be unavailable, expert root-cause annotations may be absent, and data release may be restricted by business and network-operation confidentiality.

This paper studies a cautious technical pipeline for this setting. We propose CET-STGL, a causal-inspired event-token spatio-temporal learning framework for alarm-driven KPI degradation prediction. The framework represents KPI histories as patch features, sparse alarms as event tokens and decayed intensities, cell/time information as positional features, and train-only alarm-degradation associations as an event influence prior. A cross-cell graph is inferred from available training statistics when physical topology is not present. The term causal-inspired is used deliberately: the graph organizes temporal association and event-influence hypotheses, but the observational data do not support confirmed causal discovery.

The study is deliberately narrower than telecom foundation-model or benchmark work. Recent telecom LLM studies and datasets target language or standards knowledge [7]–[9]; our problem is instead structured telemetry learning over a private enterprise operations panel. We do not train a telecom foundation model, do not release a public benchmark, do not claim strict causality, do not claim expert-confirmed root-

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cause validation, and do not report unrun neural baselines. Forecasting labels are real future KPI values. Degradation labels and root-cause ranking labels are weak labels derived from training-split KPI thresholds and train-only alarm-lift rules, following the broader weak-supervision view that noisy or indirect labels can be useful but should not be equated with ground truth [10], [11].

Our contributions are:

1) We formulate alarm-driven KPI degradation prediction as a multi-task learning problem over real cellular KPI/business and alarm/fault panels.
2) We design a causal-inspired event-token representation that combines KPI patch tokens, alarm event tokens, cell/time positional features, train-only event influence priors and inferred cross-cell graph context.
3) We provide a leakage-controlled experimental protocol for 1/3/6/12-hour KPI forecasting, weak degradation classification and weak root-cause candidate ranking.
4) We report calibrated findings: KPI-history baselines remain strong, event-only and Hawkes-only variants are weak standalone predictors, the runnable CET-STGL sklearn proxy is competitive on several primary-KPI forecasting metrics, and alarm-token gains are not uniform under current weak labels.

II. Related Work

A. KPI Forecasting and Degradation Prediction in Cellular Networks

Mobile and wireless networking research has studied prediction as a foundation for proactive management. Surveys on anticipatory mobile networking and deep learning in mobile networks summarize the use of context, traffic history and machine-learning models for prediction-driven optimization [1], [2]. More specifically, cellular traffic studies show that both temporal dependency and spatial correlation among base stations are important for accurate prediction [3], [4]. Operational KPI anomaly work such as Donut and OmniAnomaly further motivates learning from seasonal or multivariate KPI time series in production systems [12], [13]. Our work differs in that it combines KPI forecasting with weak degradation classification and weak alarm-candidate ranking on a private cell-time KPI/alarm panel, rather than only predicting traffic volume or detecting generic KPI anomalies.

B. Alarm Correlation and Root-Cause Analysis

Alarm correlation is a classic fault-management problem in communication networks. Jakobson and Weissman described alarm correlation as a way to reduce alarm floods and support fault diagnosis [5]. Bouloutas et al. modeled alarm correlation and fault identification in communication networks, emphasizing that a single fault can trigger many alarms and complicate localization [6]. More recent work has explored alarm-correlation rule mining for network fault management [14], while log-based operational diagnosis methods such as DeepLog illustrate the broader role of event sequences in detecting and diagnosing operational anomalies [15]. CET-STGL uses alarm information more conservatively: alarms provide sparse event tokens and weak ranking candidates, but the current data do not contain expert-confirmed root-cause labels.

C. Event Influence Modeling with Hawkes and Point Processes

Hawkes processes provide a classical framework for self-exciting and mutually exciting event sequences [16]. Point-process residual analysis and related statistical tools have been used to assess event-time models [17]. Modern work connects Hawkes processes to Granger-causality-style event influence learning [18], and neural point-process models extend event-intensity modeling with recurrent representations [19]. In this paper, the event module is intentionally simpler: we use Hawkes-inspired decayed counts and train-only alarm-degradation lift as event influence features. These features are causal-inspired association priors, not fitted multi-type Hawkes processes and not proof of causality.

D. Spatio-Temporal Graph Learning for Networked Time Series

Spatio-temporal graph neural networks model temporal dynamics together with dependencies among entities. DCRNN uses diffusion convolution with recurrent forecasting [20], STGCN combines graph convolution and temporal convolution for traffic forecasting [21], and Graph WaveNet learns adaptive spatial dependencies for graph time series [22]. Graph attention networks provide a general attention mechanism over graph-structured data [23]. These methods motivate the full CET-STGL design, especially the use of cross-cell context and dynamic event influence. However, the current reported experiments use a sklearn proxy because the PyTorch backend required for neural graph baselines is unavailable; no neural baseline metrics are fabricated.

E. Telecom LLMs and Benchmarks

Recent telecom LLM work studies language models for telecom knowledge, standards understanding and domain adaptation. Large-language-model impact discussions highlight possible telecom applications [7]; TeleQnA evaluates telecommunications knowledge in LLMs [8]; TelecomGPT studies domain adaptation for telecom-specific LLMs [9]; and recent benchmark proposals explore industrial telecom workflows [24]. These efforts are relevant background, but they solve a different problem. CET-STGL is not a foundation-model paper and not a benchmark paper. It is a structured-telemetry technical pipeline for KPI forecasting, weak degradation classification and weak alarm-candidate ranking on a private operations slice.

III. Problem Formulation

Let $\mathcal{C}$ be the set of cells and let t index hourly timestamps. For each cell-time pair (c, t), the panel contains a KPI/business vector $\mathbf{x}_{c,t} \in \mathbb{R}^M$, an alarm/fault vector $\mathbf{a}_{c,t} \in \mathbb{R}^A$, a cell identifier and timestamp-derived features. In the audited dataset, $|\mathcal{C}| = 33$, $M = 48$ and $A = 137$, although only 11 alarm/fault columns are active in the observed slice. Given a lookback window $\mathcal{W}_{c,t} = \{(\mathbf{x}_{c,\tau}, \mathbf{a}_{c,\tau}) : t - L + 1 \le \tau \le t\}$, the objective is to predict outcomes at horizon $h \in \{1, 3, 6, 12\}$ hours. Windows crossing the detected long time gap are removed before modeling.

A. KPI Forecasting

The forecasting task predicts future KPI values

$$\hat{\mathbf{x}}_{c,t+h} = f_h(\mathcal{W}_{c,t}, c, t). \tag{1}$$

These labels are real future observations from the private cellular panel. We report aggregate metrics over all KPI/business columns and paper-level metrics over the primary KPI subset.

B. Weak Degradation Classification

The degradation task predicts a weak binary label

$$\hat{y}_{c,t+h} = g_h(\mathcal{W}_{c,t}, c, t), \quad y_{c,t+h} \in \{0, 1\}. \tag{2}$$

The label $y_{c,t+h}$ is derived from training-split KPI thresholds, for example high call-drop rate or low throughput/radio-quality conditions. It is a weak KPI degradation label rather than an expert incident label. This distinction follows weak-supervision practice, where indirect or noisy labels can support model development but must be evaluated and interpreted as proxies [10], [11].

C. Weak Root-Cause Candidate Ranking

For degraded samples with recent alarms, the ranking task scores candidate alarm types $a \in \mathcal{A}_{c,t}$:

$$s_h(a, c, t) = r_h(a, \mathcal{W}_{c,t}, c, t). \tag{3}$$

The target candidate is induced by train-only alarm-to-degradation lift and recent event intensity. The ranking task therefore evaluates weak-label consistency only. It is not expert-confirmed root-cause localization.

D. Causal-Inspired Graph Boundary

The event influence graph is constructed from temporal precedence and train-only association. It is useful for organizing candidate event influence, but the observational data contain no interventions or expert causal annotations. We therefore use causal-inspired or causal discovery-based terminology and avoid claims of confirmed causality.

IV. Methodology

A. Overview

CET-STGL is a causal-inspired event-token spatio-temporal representation pipeline for structured cellular operations data. It contains five feature blocks: KPI patch tokens, alarm event tokens, cell/time positional features, an event influence prior and an inferred cross-cell graph. A full neural CET-STGL implementation would use these blocks in a spatio-temporal graph encoder with forecasting, classification and ranking heads. The present experiments report a runnable sklearn proxy over the same blocks because the project runtime does not include PyTorch. This proxy validates the data protocol, representation design and ablations without implying that the neural graph backend has been executed.

B. KPI Patch Tokens

For each KPI m in a lookback window, the sequence $x^{(m)}_{c,t-L+1:t}$ is divided into short patches. Each patch is summarized by simple local statistics:

$$\phi^{\text{kpi}}_{c,t,p,m} = [\text{mean}, \text{std}, \min, \max, \text{slope}]_{c,t,p,m}. \tag{4}$$

These patch summaries preserve local temporal level and trend while reducing raw sequence length. They also make the proxy robust to the multi-column KPI/business schema because all KPI columns can be transformed through the same feature construction rule.

C. Alarm Event Tokens and Intensity Features

Alarm/fault observations are represented as sparse event features. For alarm type a, we compute rolling counts, recency and a decayed event intensity

$$\lambda_a(c, t) = \sum_{\tau \le t} \mathbf{1}[a_{c,\tau} > 0] \exp\left(-\frac{t - \tau}{\eta}\right), \tag{5}$$

where η is a fixed decay scale in the feature implementation. This is Hawkes-inspired because recent events contribute more strongly than old events [16], [19], but it is not a fitted multi-type Hawkes process. The distinction matters because the current alarm observations are extremely sparse: only 11 of 137 alarm/fault columns are active and only 1.62% of rows contain an active alarm.

D. Cell and Time Positional Features

Cell identity and time context are encoded through cell-code features and cyclical time features such as hour-of-day and day-of-week. These features capture persistent cell offsets and periodic operational behavior. All transformations are fitted within the chronological training protocol, and windows crossing the detected long time gap are excluded.

E. Train-Only Event Influence Prior

For each alarm type and horizon, the proxy estimates a train-only lagged lift:

$$\text{lift}_h(a) = \log \frac{P_{\text{train}}(y_{c,t+h} = 1 \mid a \in \mathcal{W}_{c,t}) + \epsilon}{P_{\text{train}}(y_{c,t+h} = 1) + \epsilon}. \tag{6}$$

The lift is combined with recent intensity to create event influence features. This is an event influence prior: it encodes temporal association between alarms and weak degradation under training data only. It is causal-inspired and leakage-controlled, but it is not evidence of confirmed causal effect.

F. Inferred Cross-Cell Graph

The audited data do not include verified physical topology. We therefore infer a cross-cell context graph from training-period statistics, such as KPI correlation. Let $G_{\text{cell}} = (\mathcal{C}, E)$ denote the inferred graph. Neighbor features summarize KPI context from cells linked to c in G_{cell}. This graph is a statistical context graph, not a physical deployment graph.

G. Prediction Heads and Proxy Implementation

The runnable proxy uses sklearn-compatible heads: Ridge regression for multi-output KPI forecasting, logistic-style classification for weak degradation, and event-lift/intensity scoring for weak ranking. The intended full CET-STGL design would replace these heads with a neural spatio-temporal graph encoder and task-specific projections, conceptually aligned with spatio-temporal graph models [20]–[22]. However, because PyTorch is unavailable in the current runtime, LSTM, TCN, Informer, Autoformer, STGCN and GAT are explicitly marked as `not_run_backend_missing`. No neural metrics are reported or inferred.

V. Experimental Setup

A. Dataset

The private panel contains 103,983 rows, 187 original columns, 33 cells and 3,151 unique hourly timestamps from 2024-01-10 00:00:00 to 2024-06-10 06:00:00. It includes 48 KPI/business columns and 137 alarm/fault columns. No missing values were detected. One long gap of 21 days and 1 hour is present, and crossing windows are removed. Table I summarizes the data scale and sparsity.

B. Splits, Labels and Horizons

All experiments use chronological splits by timestamp: 70% train, 15% validation and 15% test. All scalers, weak thresholds, inferred graphs and event influence scores are fitted from the training split only. Horizons are 1, 3, 6 and 12 hours. Forecasting labels are real future KPI values. Degradation labels are weak KPI-threshold labels, and ranking labels are weak alarm-lift/event-intensity candidates. These weak labels are useful for protocol development, but they should not be read as expert ground truth [10].

C. Compared Models

Runnable models include Ridge/Logistic, Ridge/Logistic without alarms, Hawkes-only, Event-attention-only, CET-STGL-sklearn and CET ablations removing alarm tokens, dynamic event graph, Hawkes/event intensity, tokenization and cross-cell graph. Appendix Table X reports neural backend status. LSTM, TCN, Informer, Autoformer, STGCN and GAT are not run because PyTorch is unavailable.

D. Metrics

Forecasting uses MAE, RMSE and MAPE with denominator floor $\max(|y|, 1.0)$. Weak degradation classification uses AUPRC, AUROC, F1, precision and recall. Weak ranking uses Hit@1, Hit@3, Hit@5, MRR and nDCG@5, interpreted as weak-label consistency only.

VI. Results and Analysis

A. KPI Forecasting

Table II and Fig. 1 summarize primary-KPI forecasting. CET-STGL-sklearn is competitive with the Ridge/Logistic baseline on several horizons. At 1 hour, CET-STGL-sklearn obtains MAE/RMSE 2.183/4.635, compared with 2.219/4.698 for Ridge/Logistic. At 3 hours the corresponding values are 2.537/5.083 versus 3.071/5.684. At 6 hours they are 3.197/5.853 versus 3.338/5.940. At 12 hours CET-STGL-sklearn has slightly lower MAE but slightly higher RMSE than Ridge/Logistic.

This evidence supports a restrained claim of competitiveness rather than universal superiority. The no-alarm CET ablation has the lowest primary MAE at all four horizons in the current results. Therefore, the paper cannot claim that alarm tokens uniformly improve forecasting. The likely explanation is data-driven: active alarms occur in only 1.62% of rows, while many KPI values are governed by temporal demand, load and radio-quality dynamics.

Event-only methods are much weaker for continuous forecasting. Hawkes-only has 1-hour primary MAE/RMSE 4.578/7.981, and Event-attention-only is similarly weak. Sparse alarm events alone therefore do not reconstruct continuous KPI trajectories in this slice.

B. Weak KPI Degradation Classification

Table III and Fig. 2 report weak degradation classification. At 1 hour, CET-STGL-sklearn obtains AUPRC/F1 0.466/0.479. Ridge/Logistic obtains 0.471/0.477, while Ridge/Logistic without alarms obtains 0.476/0.480. At later horizons, the relative ordering changes by metric and horizon, but the overall message is stable: alarm/event features do not provide a uniform gain under weak KPI-threshold labels.

The event-only baselines again expose the sparsity limitation. Hawkes-only and Event-attention-only are weaker than KPI-history baselines in AUPRC at the 1-hour horizon, with AUPRC 0.247 and 0.257 respectively. This does not mean alarms are useless; it means that, in this dataset, alarms are better interpreted as sparse event context and weak ranking candidates than as standalone predictors of threshold-derived degradation.

C. Weak Root-Cause Candidate Ranking

Table IV and Fig. 4 show weak ranking consistency. CET-STGL-sklearn reaches perfect weak-label consistency in this setup, while Hawkes-only obtains Hit@1 around 0.80 and MRR around 0.89 across horizons. These numbers must not be interpreted as true root-cause localization accuracy. The ranking labels are derived from train-only alarm-to-degradation lift and recent event intensity, and some model variants use overlapping signals in scoring. The appropriate interpretation is that the ranking head can recover proxy candidates induced by the weak-label rule.

D. Ablation Analysis

Table V and Fig. 3 summarize ablation deltas relative to CET-STGL-sklearn. Removing alarm tokens improves average forecasting MAE relative to the full proxy, which is consistent with the finding that KPI history dominates aggregate forecasting in this sparse-alarm slice. Removing Hawkes/event intensity changes several metrics only slightly, suggesting that

TABLE I
DATASET STATISTICS FOR THE PRIVATE CELLULAR OPERATIONS PANEL.

Item	Value
Rows	103,983
Original columns	187
Cells	33
Unique hourly timestamps	3,151
Time range	2024-01-10 00:00:00 to 2024-06-10 06:00:00
KPI/business columns	48
Alarm/fault columns	137
Active alarm/fault columns	11
Rows with active alarms	1,681 (1.62%)
Missing values	None detected
Detected time discontinuity	One 21 days 01:00:00 gap; crossing windows removed

Note. The data are a private cellular operations slice from `510957.csv`, not a public benchmark. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

TABLE II
PRIMARY-KPI FORECASTING SUMMARY FOR RUNNABLE MODELS.

Model	H	MAE	RMSE	MAPE
CET w/o alarm tokens	1	2.157	4.602	42.411
CET-STGL-sklearn	1	2.183	4.635	42.642
Event-attention-only	1	4.630	8.276	82.519
Hawkes-only	1	4.578	7.981	82.261
Ridge/Logistic	1	2.219	4.698	43.599
Ridge/Logistic w/o alarms	1	2.203	4.671	43.382
CET w/o alarm tokens	3	2.490	4.974	46.807
CET-STGL-sklearn	3	2.537	5.083	47.162
Event-attention-only	3	4.635	8.326	82.776
Hawkes-only	3	4.581	8.038	82.515
Ridge/Logistic	3	3.071	5.684	56.382
Ridge/Logistic w/o alarms	3	3.036	5.586	56.015
CET w/o alarm tokens	6	3.145	5.605	56.843
CET-STGL-sklearn	6	3.197	5.853	57.264
Event-attention-only	6	4.629	8.277	83.117
Hawkes-only	6	4.578	7.873	82.871
Ridge/Logistic	6	3.338	5.940	61.590
Ridge/Logistic w/o alarms	6	3.318	5.884	61.272
CET w/o alarm tokens	12	2.515	5.051	47.283
CET-STGL-sklearn	12	2.604	5.442	47.682
Event-attention-only	12	4.671	8.622	83.341
Hawkes-only	12	4.572	7.888	83.061
Ridge/Logistic	12	2.640	5.391	49.140
Ridge/Logistic w/o alarms	12	2.593	5.135	48.806

Note. Forecasting targets are real future KPI values. MAPE uses denominator floor $\max(|y|, 1.0)$. Neural baselines are not run because PyTorch is unavailable. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

TABLE III
WEAK KPI DEGRADATION CLASSIFICATION SUMMARY.

Model	H	AUPRC	F1	Prec.	Recall
CET w/o alarm tokens	1	0.470	0.361	0.221	0.975
CET-STGL-sklearn	1	0.466	0.479	0.395	0.608
Event-attention-only	1	0.257	0.331	0.226	0.620
Hawkes-only	1	0.247	0.327	0.218	0.656
Ridge/Logistic	1	0.471	0.477	0.418	0.556
Ridge/Logistic w/o alarms	1	0.476	0.480	0.396	0.610
CET w/o alarm tokens	3	0.399	0.348	0.520	0.262
CET-STGL-sklearn	3	0.375	0.147	0.570	0.084
Event-attention-only	3	0.249	0.331	0.218	0.684
Hawkes-only	3	0.233	0.326	0.218	0.649
Ridge/Logistic	3	0.341	0.320	0.191	0.996
Ridge/Logistic w/o alarms	3	0.336	0.212	0.445	0.139
CET w/o alarm tokens	6	0.334	0.331	0.200	0.975
CET-STGL-sklearn	6	0.318	0.389	0.284	0.618
Event-attention-only	6	0.215	0.352	0.245	0.623
Hawkes-only	6	0.294	0.318	0.215	0.611
Ridge/Logistic	6	0.309	0.315	0.402	0.259
Ridge/Logistic w/o alarms	6	0.284	0.363	0.245	0.698
CET w/o alarm tokens	12	0.377	0.334	0.201	0.989
CET-STGL-sklearn	12	0.373	0.324	0.193	0.997
Event-attention-only	12	0.240	0.340	0.238	0.590
Hawkes-only	12	0.247	0.322	0.219	0.608
Ridge/Logistic	12	0.381	0.334	0.201	0.983
Ridge/Logistic w/o alarms	12	0.396	0.335	0.202	0.983

Note. Labels are weak KPI-threshold labels fitted from the training split, not expert incident labels. Neural baselines are not run because PyTorch is unavailable. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

current decayed-intensity features are either redundant with simpler alarm counts or too sparse to dominate. Removing tokenization has a larger adverse forecasting effect, indicating that patch-style KPI summaries are useful for the current proxy. Removing the dynamic event graph hurts weak ranking consistency because the ranking proxy depends on alarm-lift structure; this is a weak-label diagnostic, not root-cause proof.

VII. DISCUSSION AND LIMITATIONS

The experiments support a technical but bounded conclusion. CET-STGL is useful as a representation framework for combining KPI history, sparse alarms, cell/time context and inferred graph information. The runnable sklearn proxy is competitive with Ridge/Logistic on several primary-KPI forecasting metrics, and the ablations show which representation blocks are active in the current data. At the same time, the results do not support broad claims of alarm-token superiority or broad neural-graph superiority.

The main empirical lesson is that KPI history is strong. This is expected for hourly KPI and business panels with regular load and radio patterns. Alarm features are operationally meaningful, but they are sparse in this slice. Therefore, event-only and Hawkes-only variants are weak standalone predictors. Their weakness strengthens the methodological point: alarms should be integrated with KPI context and treated as event-conditioned evidence, not used alone to explain KPI degradation.

The root-cause ranking task is intentionally framed as weak-label consistency. High Hit@K or MRR values indicate agreement with train-only proxy labels, not expert-confirmed root-cause localization. This is especially important because some ranking scores use signals related to the weak-label

TABLE IV
WEAK ROOT-CAUSE CANDIDATE RANKING SUMMARY.

Model	H	N	Hit@1	MRR	nDCG@5
CET w/o dynamic event graph	1	2682	0.808	0.895	0.920
CET-STGL-sklearn	1	2682	1.000	1.000	1.000
Event-attention-only	1	611	1.000	1.000	1.000
Hawkes-only	1	2682	0.808	0.895	0.920
Ridge/Logistic	1	611	0.930	0.960	0.971
CET w/o dynamic event graph	3	2671	0.807	0.894	0.919
CET-STGL-sklearn	3	2671	1.000	1.000	1.000
Event-attention-only	3	609	1.000	1.000	1.000
Hawkes-only	3	2671	0.807	0.894	0.919
Ridge/Logistic	3	609	0.923	0.958	0.969
CET w/o dynamic event graph	6	2645	0.805	0.893	0.918
CET-STGL-sklearn	6	2645	1.000	1.000	1.000
Event-attention-only	6	614	1.000	1.000	1.000
Hawkes-only	6	2645	0.805	0.893	0.918
Ridge/Logistic	6	614	0.917	0.955	0.967
CET w/o dynamic event graph	12	2617	0.804	0.892	0.918
CET-STGL-sklearn	12	2617	1.000	1.000	1.000
Event-attention-only	12	649	1.000	1.000	1.000
Hawkes-only	12	2617	0.804	0.892	0.918
Ridge/Logistic	12	649	0.928	0.962	0.972

Note. Ranking uses weak alarm-lift/event-intensity proxy labels, not expert-confirmed root-cause localization. Perfect scores indicate proxy consistency when scoring overlaps with weak-label construction. Neural baselines are not run because PyTorch is unavailable. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

TABLE V
ABLATION DELTAS AVERAGED OVER HORIZONS RELATIVE TO CET-STGL-SKLEARN.

Ablation row	Removed component	Δ MAE	Δ AUPRC	Δ MRR
CET w/o alarm tokens	no_alarm	-0.054	0.012	
CET w/o cross-cell graph	no_cross_cell_graph	-0.003	-0.000	0.000
CET w/o dynamic event graph	no_dynamic_graph	0.000	0.001	-0.107
CET w/o Hawkes intensity	no_hawkes_intensity	0.000	0.001	0.000
CET w/o tokenization	no_tokenization	0.200	-0.017	0.000

Note. Negative MAE deltas are better; positive AUPRC/MRR deltas are better. Deltas are derived from existing paper artifacts only. Neural baselines are not run because PyTorch is unavailable. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

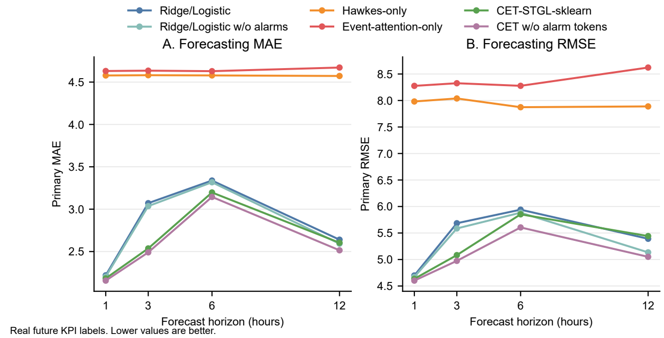

Fig. 1. Primary-KPI forecasting error versus horizon. The plot uses real future KPI values from the private panel. Lower values are better. Neural baselines are not included because PyTorch is unavailable.

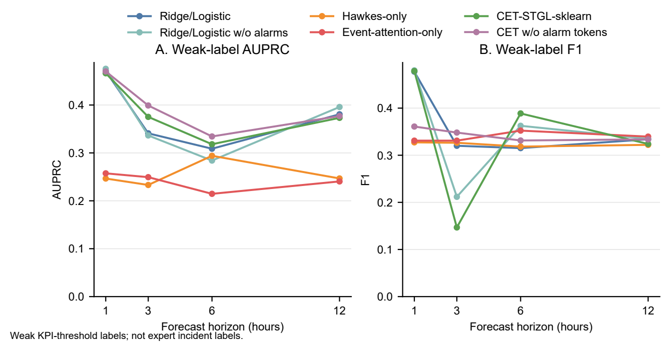

Fig. 2. Weak KPI degradation classification under training-split threshold labels. Scores are not expert incident-detection accuracy, and alarm/event features do not uniformly improve performance under the current weak labels.

construction. Expert incident tickets, work orders or validated root-cause annotations would be required for true operational RCA evaluation.

Several limitations remain. First, the data are private and come from one project slice, so the work is not a public benchmark and does not establish broad generalization across operators or regions. Second, degradation and ranking labels are weak labels; only forecasting uses real future KPI targets. Third, the event influence graph is causal-inspired rather than strictly causal. Observational associations cannot establish causality without interventions, counterfactual assumptions or external validation. Fourth, neural baselines were not run because PyTorch is unavailable; LSTM, TCN, Informer, Autoformer, STGCN and GAT remain future implementation targets. Fifth, the cross-cell graph is inferred statistically rather than verified from physical topology.

VIII. DATA AVAILABILITY AND REPRODUCIBILITY

The raw dataset is a private cellular operational record and cannot be released as a public benchmark in its current form. It contains business-sensitive KPI and alarm information and may reveal network-operation characteristics. Public release would require enterprise approval, anonymization review and possibly synthetic or schema-only alternatives.

The reproducible components of this project are the experimental protocol, code skeleton, feature-construction logic, model-status reporting and table/figure generation scripts. If allowed by the data owner, an anonymized schema, toy example or synthetic panel can be released in future work to demonstrate the pipeline without exposing operational records. Until then, all reported metrics should be interpreted as results on the private `510957.csv` slice, and not as a public benchmark leaderboard.

IX. CONCLUSION

This paper presented CET-STGL, a causal-inspired event-token spatio-temporal learning pipeline for alarm-driven KPI degradation prediction in cellular networks. On a real private cellular operations panel, the framework structures KPI histories, sparse alarms, cell/time context and inferred graph information for three tasks: multi-horizon KPI forecasting, weak KPI degradation classification and weak root-cause candidate ranking.

The current evidence supports a calibrated technical claim. The runnable sklearn proxy is competitive with Ridge/Logistic baselines on several primary-KPI forecasting metrics, while event-only and Hawkes-only variants are weak standalone predictors under sparse alarm observations. Ablations show

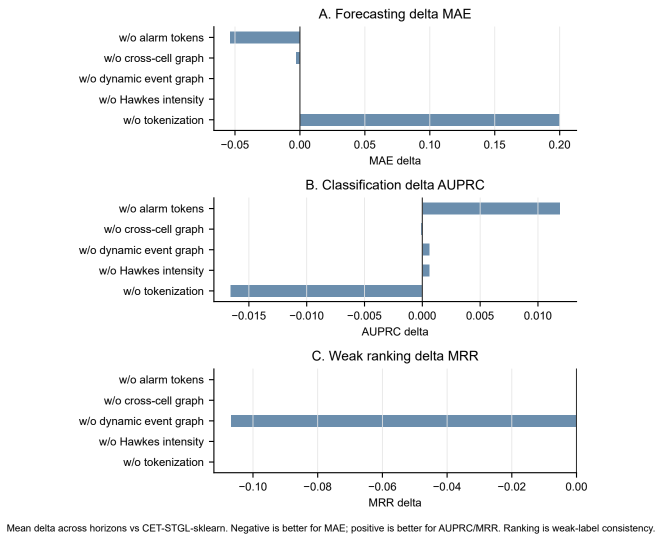

Fig. 3. Ablation deltas relative to CET-STGL-sklearn. Negative deltas are better for MAE, while positive deltas are better for AUPRC and MRR. Ranking deltas reflect weak-label/proxy consistency only.

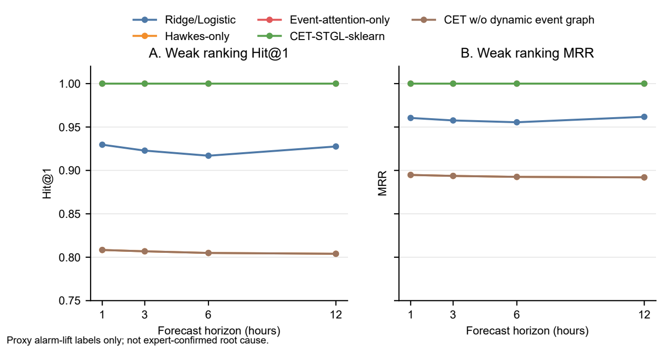

Fig. 4. Weak root-cause candidate ranking consistency. These are proxy labels derived from train-only alarm lift and recent event intensity, not expert-confirmed root causes.

that no-alarm variants can be competitive or better in several settings, so alarm tokens should not be claimed to uniformly improve performance. Weak ranking results should be interpreted only as proxy consistency, not true root-cause validation.

Future work should add expert incident/root-cause annotations, evaluate additional regions or time periods, incorporate verified topology where available and implement the full neural spatio-temporal graph encoder under the same leakage-controlled protocol.

APPENDIX A
FULL EXPERIMENTAL TABLES

The main paper reports compact summary tables for readability. This appendix retains the full runnable model-by-horizon tables generated from the existing paper artifacts. Neural baselines are listed only in the backend-status table; no neural metrics are fabricated.

A. Full Forecasting Results

TABLE VI: KPI forecasting results over runnable models and horizons.

Model	H	P-MAE	P-RMSE	P-MAPE	All-MAE	All-RMSE
CET w/o alarm tokens	1	2.1569	4.6016	42.4111	30.1717	53.1923
CET w/o cross-cell graph	1	2.1791	4.6351	42.7362	31.1928	62.4168
CET w/o dynamic event graph	1	2.1831	4.6351	42.6425	31.2506	62.1414
CET w/o Hawkes intensity	1	2.1831	4.6351	42.6425	31.2506	62.1414
CET w/o tokenization	1	2.2244	4.7059	43.4812	31.6755	61.3319
CET-STGL-sklearn	1	2.1830	4.6351	42.6425	31.2510	62.1427
Event-attention-only	1	4.6305	8.2757	82.5194	163.9535	271.1688
Hawkes-only	1	4.5779	7.9814	82.2610	160.0211	227.2084
Ridge/Logistic	1	2.2191	4.6984	43.5993	31.4426	58.4626
Ridge/Logistic w/o alarms	1	2.2034	4.6712	43.3820	30.6931	53.7885
CET w/o alarm tokens	3	2.4904	4.9744	46.8069	39.4850	67.8885
CET w/o cross-cell graph	3	2.5354	5.0786	47.2707	42.1875	103.4692
CET w/o dynamic event graph	3	2.5368	5.0829	47.1621	42.1587	102.7291
CET w/o Hawkes intensity	3	2.5368	5.0829	47.1621	42.1587	102.7291
CET w/o tokenization	3	3.0549	5.6840	56.1056	50.7988	101.8653
CET-STGL-sklearn	3	2.5367	5.0829	47.1624	42.1585	102.7298
Event-attention-only	3	4.6346	8.3256	82.7764	164.6453	289.4013
Hawkes-only	3	4.5809	8.0385	82.5149	159.7122	227.2520
Ridge/Logistic	3	3.0709	5.6842	56.3815	50.7191	93.6261
Ridge/Logistic w/o alarms	3	3.0361	5.5858	56.0146	48.9354	78.7454
CET w/o alarm tokens	6	3.1447	5.6052	56.8431	50.7871	82.1649
CET w/o cross-cell graph	6	3.1904	5.8644	57.2840	54.4509	133.8707
CET w/o dynamic event graph	6	3.1970	5.8527	57.2636	54.4014	132.2650
CET w/o Hawkes intensity	6	3.1970	5.8527	57.2636	54.4014	132.2650
CET w/o tokenization	6	3.3631	6.0530	61.5916	65.4517	131.9830
CET-STGL-sklearn	6	3.1971	5.8527	57.2638	54.4004	132.2636
Event-attention-only	6	4.6287	8.2769	83.1170	164.8499	295.9310
Hawkes-only	6	4.5785	7.8735	82.8707	159.5017	225.5720
Ridge/Logistic	6	3.3380	5.9402	61.5904	65.1005	118.3171
Ridge/Logistic w/o alarms	6	3.3177	5.8838	61.2722	62.6208	97.5729
CET w/o alarm tokens	12	2.5147	5.0508	47.2830	42.0699	74.8636
CET w/o cross-cell graph	12	2.6040	5.4528	47.7833	46.2547	127.1682
CET w/o dynamic event graph	12	2.6041	5.4421	47.6820	46.4562	126.0803
CET w/o Hawkes intensity	12	2.6041	5.4421	47.6820	46.4562	126.0803
CET w/o tokenization	12	2.6796	5.5472	49.0619	47.8638	124.7888
CET-STGL-sklearn	12	2.6041	5.4421	47.6819	46.4562	126.0802
Event-attention-only	12	4.6714	8.6216	83.3406	164.9266	287.0291
Hawkes-only	12	4.5719	7.8884	83.0608	160.6519	238.1456
Ridge/Logistic	12	2.6400	5.3913	49.1403	47.3654	122.6165
Ridge/Logistic w/o alarms	12	2.5928	5.1347	48.8058	43.2913	76.8817

Note. Forecasting labels are real future KPI values. MAPE uses denominator floor $\max(|y|, 1.0)$. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

B. Full Weak Degradation Classification Results

TABLE VII: Weak KPI degradation classification results.

Model	H	AUPRC	F1	Prec.	Recall	AUROC
CET w/o alarm tokens	1	0.4702	0.3609	0.2215	0.9746	0.7975
CET w/o cross-cell graph	1	0.4482	0.3980	0.5244	0.3207	0.7827
CET w/o dynamic event graph	1	0.4662	0.4784	0.3948	0.6070	0.7943
CET w/o Hawkes intensity	1	0.4662	0.4785	0.3947	0.6074	0.7944
CET w/o tokenization	1	0.4578	0.4583	0.4634	0.4534	0.7892
CET-STGL-sklearn	1	0.4663	0.4787	0.3949	0.6078	0.7944
Event-attention-only	1	0.2573	0.3310	0.2257	0.6201	0.6358
Hawkes-only	1	0.2465	0.3274	0.2181	0.6562	0.6314
Ridge/Logistic	1	0.4709	0.4774	0.4182	0.5563	0.7946
Ridge/Logistic w/o alarms	1	0.4756	0.4801	0.3956	0.6104	0.7971
CET w/o alarm tokens	3	0.3990	0.3481	0.5197	0.2617	0.7587
CET w/o cross-cell graph	3	0.3870	0.4010	0.2700	0.7787	0.7527
CET w/o dynamic event graph	3	0.3760	0.1475	0.5751	0.0846	0.7430
CET w/o Hawkes intensity	3	0.3759	0.1476	0.5780	0.0846	0.7429
CET w/o tokenization	3	0.3419	0.3121	0.1849	0.9985	0.7236

Model	H	AUPRC	F1	Prec.	Recall	AUROC
CET-STGL-sklearn	3	0.3752	0.1468	0.5696	0.0842	0.7427
Event-attention-only	3	0.2494	0.3310	0.2183	0.6836	0.6328
Hawkes-only	3	0.2331	0.3265	0.2181	0.6492	0.6230
Ridge/Logistic	3	0.3409	0.3201	0.1907	0.9959	0.7231
Ridge/Logistic w/o alarms	3	0.3365	0.2118	0.4454	0.1389	0.7256
CET w/o alarm tokens	6	0.3344	0.3313	0.1996	0.9750	0.7167
CET w/o cross-cell graph	6	0.3195	0.3318	0.1997	0.9792	0.7178
CET w/o dynamic event graph	6	0.3203	0.3894	0.2853	0.6129	0.7131
CET w/o Hawkes intensity	6	0.3203	0.3892	0.2849	0.6136	0.7131
CET w/o tokenization	6	0.3026	0.3033	0.3729	0.2556	0.6842
CET-STGL-sklearn	6	0.3181	0.3887	0.2835	0.6181	0.7127
Event-attention-only	6	0.2146	0.3523	0.2455	0.6234	0.6165
Hawkes-only	6	0.2940	0.3183	0.2152	0.6113	0.6326
Ridge/Logistic	6	0.3087	0.3152	0.4025	0.2590	0.6909
Ridge/Logistic w/o alarms	6	0.2840	0.3625	0.2448	0.6983	0.6830
CET w/o alarm tokens	12	0.3769	0.3337	0.2007	0.9885	0.7526
CET w/o cross-cell graph	12	0.3777	0.3213	0.1915	0.9981	0.7499
CET w/o dynamic event graph	12	0.3728	0.3243	0.1937	0.9966	0.7463
CET w/o Hawkes intensity	12	0.3729	0.3243	0.1937	0.9966	0.7463
CET w/o tokenization	12	0.3641	0.3633	0.4503	0.3045	0.7382
CET-STGL-sklearn	12	0.3732	0.3240	0.1934	0.9966	0.7465
Event-attention-only	12	0.2405	0.3396	0.2383	0.5904	0.6332
Hawkes-only	12	0.2467	0.3221	0.2190	0.6083	0.6107
Ridge/Logistic	12	0.3806	0.3339	0.2011	0.9828	0.7413
Ridge/Logistic w/o alarms	12	0.3959	0.3348	0.2018	0.9828	0.7420

Note. Classification labels are weak KPI-threshold labels fitted from the training split; they are not expert incident labels. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

C. Full Weak Ranking Results

TABLE VIII: Weak root-cause candidate ranking consistency.

Model	H	N	Hit@1	Hit@3	Hit@5	MRR	nDCG@5
CET w/o alarm tokens	1	0					
CET w/o cross-cell graph	1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o dynamic event graph	1	2682	0.8084	0.9952	0.9959	0.8948	0.9204
CET w/o Hawkes intensity	1	611	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o tokenization	1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn	1	2682	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only	1	611	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only	1	2682	0.8084	0.9952	0.9959	0.8948	0.9204
Ridge/Logistic	1	611	0.9296	1.0000	1.0000	0.9604	0.9706
Ridge/Logistic w/o alarms	1	0					
CET w/o alarm tokens	3	0					
CET w/o cross-cell graph	3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o dynamic event graph	3	2671	0.8068	0.9944	0.9951	0.8937	0.9193
CET w/o Hawkes intensity	3	609	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o tokenization	3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn	3	2671	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only	3	609	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only	3	2671	0.8068	0.9944	0.9951	0.8937	0.9193
Ridge/Logistic	3	609	0.9228	1.0000	1.0000	0.9576	0.9685
Ridge/Logistic w/o alarms	3	0					
CET w/o alarm tokens	6	0					
CET w/o cross-cell graph	6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o dynamic event graph	6	2645	0.8049	0.9940	0.9947	0.8926	0.9183
CET w/o Hawkes intensity	6	614	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o tokenization	6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn	6	2645	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only	6	614	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only	6	2645	0.8049	0.9940	0.9947	0.8926	0.9183
Ridge/Logistic	6	614	0.9169	1.0000	1.0000	0.9555	0.9670
Ridge/Logistic w/o alarms	6	0					
CET w/o alarm tokens	12	0					
CET w/o cross-cell graph	12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o dynamic event graph	12	2617	0.8040	0.9935	0.9943	0.8920	0.9177
CET w/o Hawkes intensity	12	649	1.0000	1.0000	1.0000	1.0000	1.0000
CET w/o tokenization	12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
CET-STGL-sklearn	12	2617	1.0000	1.0000	1.0000	1.0000	1.0000
Event-attention-only	12	649	1.0000	1.0000	1.0000	1.0000	1.0000
Hawkes-only	12	2617	0.8040	0.9935	0.9943	0.8920	0.9177
Ridge/Logistic	12	649	0.9276	1.0000	1.0000	0.9617	0.9717
Ridge/Logistic w/o alarms	12	0					

Note. Weak root-cause ranking uses train-only alarm-lift/event-intensity proxy labels; it is not expert-confirmed root-cause localization. Perfect scores indicate weak-label consistency when the scoring rule overlaps with the weak-label construction. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

TABLE X
NEURAL BASELINE BACKEND STATUS.

Model	Status	Reason
lstm	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.
tcn	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.
informer	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.
autoformer	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.
stgcn	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.
gat	not_run backend_missing	PyTorch is not installed in the current runtime; sklearn proxy and interfaces are runnable.

Note. No metrics are reported for these neural baselines. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

D. Full Ablation Delta Table

TABLE IX: Ablation deltas relative to CET-STGL-sklearn.

Ablation row	Removed	H	Δ P-MAE	Δ AUPRC	Δ MRR
CET w/o alarm tokens	no_alarm	1	-0.0262	0.0039	
CET w/o alarm tokens	no_alarm	3	-0.0463	0.0238	
CET w/o alarm tokens	no_alarm	6	-0.0524	0.0162	
CET w/o alarm tokens	no_alarm	12	-0.0893	0.0037	
CET w/o cross-cell graph	no_cross_cell_graph	1	-0.0039	-0.0181	0.0000
CET w/o cross-cell graph	no_cross_cell_graph	3	-0.0014	0.0118	0.0000
CET w/o cross-cell graph	no_cross_cell_graph	6	-0.0066	0.0014	0.0000
CET w/o cross-cell graph	no_cross_cell_graph	12	-0.0001	0.0045	0.0000
CET w/o dynamic event graph	no_dynamic_graph	1	0.0001	-0.0001	-0.1052
CET w/o dynamic event graph	no_dynamic_graph	3	0.0000	0.0007	-0.1063
CET w/o dynamic event graph	no_dynamic_graph	6	-0.0001	0.0022	-0.1074
CET w/o dynamic event graph	no_dynamic_graph	12	-0.0000	-0.0004	-0.1080
CET w/o Hawkes intensity	no_hawkes_intensity	1	0.0001	-0.0001	0.0000
CET w/o Hawkes intensity	no_hawkes_intensity	3	0.0000	0.0007	0.0000
CET w/o Hawkes intensity	no_hawkes_intensity	6	-0.0001	0.0022	0.0000
CET w/o Hawkes intensity	no_hawkes_intensity	12	-0.0000	-0.0003	0.0000
CET w/o tokenization	no_tokenization	1	0.0414	-0.0085	0.0000
CET w/o tokenization	no_tokenization	3	0.5181	-0.0333	0.0000
CET w/o tokenization	no_tokenization	6	0.1660	-0.0155	0.0000
CET w/o tokenization	no_tokenization	12	0.0755	-0.0091	0.0000

Note. Deltas are computed against CET-STGL-sklearn from existing paper artifacts. Negative MAE deltas are better; positive AUPRC/MRR deltas are better. Neural baselines are not run because the PyTorch backend is unavailable. Root-cause ranking is weak-label/proxy consistency, not expert-confirmed root cause. The event graph is causal-inspired / causal discovery-based, not confirmed causality.

E. Backend Status

APPENDIX B
ADDITIONAL REPRODUCIBILITY NOTES

All reported runnable metrics are derived from the private `510957.csv` panel or weak labels fitted from the training split. Forecasting uses real future KPI values. Weak degradation classification uses KPI-threshold labels, and weak root-cause ranking uses train-only alarm-lift/event-intensity candidates. The table and figure source CSV/JSON files are retained in `paper_cet_stgl/source_artifacts/`. The script `scripts/build_cet_stgl_paper_package.py` records the earlier table/figure generation process; the current manuscript does not rerun any model.

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